

Task 8: Build a Functional To-Do List Application (CRUD)

Tools:

- Primary: VS Code, Browser (Chrome/Edge), JavaScript (Vanilla)
- Alternatives: CodePen, StackBlitz, JSFiddle

Hints / Mini Guide:

1. Create a basic HTML structure with an input field, add button, and an empty list container where tasks will be dynamically rendered.
2. Style the layout using CSS to clearly differentiate input area, task list, completed tasks, and action buttons to make the UI user-friendly.
3. Write JavaScript to capture user input when the add button is clicked or when Enter key is pressed.
4. Dynamically create list items using `document.createElement()` instead of writing HTML strings to follow safer DOM manipulation practices.
5. Add functionality to mark tasks as completed by toggling a CSS class using JavaScript event listeners.
6. Implement delete functionality for each task using event delegation to handle dynamically added elements efficiently.
7. Store tasks in an array and sync it with the UI to understand state management fundamentals.
8. Add basic validation to prevent empty tasks from being added.
9. Continuously test the application in the browser and debug issues using console logs and DevTools.

Deliverables:

- Fully functional to-do list app with add, delete, and complete features
- Clean DOM manipulation without page reloads

Interview Questions Related To Above Task:

- What is CRUD and how is it applied here?
- Why is event delegation important?
- Difference between `innerHTML` and `createElement()`?
- How do you manage state in JavaScript?
- What are JavaScript event listeners?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

